

A Formal Companion to *A Database of Rigorous Maass Forms*

A machine-checked Lean 4 development: automorphic setup, cusp-width theory, certification cores, and the arithmetic of the announced snapshot

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Abstract

This paper documents a Lean 4 / Mathlib development that formalizes a mathematical companion to David Lowry-Duda’s database announcement *A Database of Rigorous Maass Forms* (arXiv:2502.01442v1). The development consists of 82 modules and 14,728 lines of Lean containing 662 machine-checked theorems and lemmas and no `sorry`. We describe what it proves: the $\mathrm{SL}_2(\mathbb{R})$ action on the upper half-plane and the hyperbolic Laplacian, the spectral relation $\lambda = \frac{1}{4} + r^2$, separation of variables into the modified-Bessel radial equation, the $\Gamma_0(N)$ automorphic and cuspidality package with fundamental covers and orbit quotients, a complete theory of exact cusp widths $w(\sigma) = N/\mathrm{gcd}(N, c^2)$ with invariance, classification and a geometric $\mathbb{P}^1(\mathbb{Q})$ model, finite/abstract cores of the three certification algorithms used by the database (trace-formula isolation, quasimode certification, Hejhal horocycle extraction with contraction certificates), a sound center-error interval calculus with Fricke-sign detection, and the exact combinatorics of the announced snapshot—65 squarefree levels, 231 level-tagged cusp classes, their width distribution, a complementary-width duality with 116 orbits and the structural decomposition $231 = 1 + 2 \cdot 115$. Twelve diagrams visualize the formal content. We are explicit about what is *not* formalized: no analytic K_{ir} theory, no Selberg trace formula, no executable certified Hejhal solver, and no ingestion of the announced records; every numerical constant of the announcement is an input, so theorems about it check arithmetic, not empirical truth.

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1 Introduction

1.1 The source and the object of study

The primary source of this project is the five-page database announcement *A Database of Rigorous Maass Forms* by David Lowry-Duda (arXiv:2502.01442v1, 3 February 2025). It announces a collection of 35,416 rigorously certified Maass cusp forms for $\Gamma_0(N)$ at squarefree levels $1 \leq N \leq 105$, obtained by combining three certified computational methods: a rigorous trace formula, a quasimode certification argument, and a rigorous version of Hejhal’s algorithm. The announcement contains no proofs; it sets up standard mathematics, sketches three algorithms, and reports meta-data about the resulting database.

The object described in this paper is different: it is a *formal companion*, a Lean4 library (`RequestProject/`) in which the mathematical skeleton around that announcement is stated and

proved by machine. The library is what we call, throughout, *the proof system*: a set of definitions in the Calculus of Inductive Constructions, together with kernel-checked proofs, built on Mathlib.

1.2 What this paper is

This document is a report on that development. Each mathematical statement below is attributed to the Lean declaration that proves it: a theorem displayed here is a human-readable rendering of a formal statement, and the declaration name in the margin note is the ground truth. Nothing in this paper is asserted mathematically unless the corresponding Lean declaration compiles without **sorry**. Conversely, and just as importantly, Section 9 lists in detail what the development does *not* prove.

1.3 Summary of the development

Measured quantity	Value
Lean source files (<code>RequestProject/*.lean</code>)	82
Lines of Lean	14,728
theorem declarations	651
lemma declarations	11
def declarations (incl. noncomputable)	380
structure / abbrev / instance	23 / 19 / 7
Occurrences of sorry or admit	0
Files using native_decide	25
lake build result	success, 8108 jobs, 0 errors
Toolchain	Lean 4.28.0, Mathlib 8f9d9cff6bd7

Table 1: Size and build status of the development, measured on the state of the repository described here.

1.4 Structure of the library

Figure 1 shows the layered structure. The lowest layer is the analytic core (Section 3); above it the automorphic and fundamental-domain layer (Section 4); above that the cusp theory (Section 5), which is the mathematically deepest part of the development; in parallel, the certification cores (Section 6) formalizing finite and abstract analogues of the three algorithms; and on top the database arithmetic (Section 7), where cusp theory and metadata meet. A sixth group of modules (Section 8) contains verified mathematics that goes beyond the announcement.

1.5 Contributions

1. A complete, sorry-free formalization of the elementary analytic setup of the announcement: Möbius action, hyperbolic Laplacian, the spectral relation, separation of variables, and the passage from a solution of the modified Bessel equation to a genuine Laplace eigenfunction (Section 3).
2. A formal $\Gamma_0(N)$ automorphic package with cuspidality at every cusp, a finite fundamental cover at every level, and an exact set-theoretic orbit transversal (Section 4).
3. A complete theory of *exact cusp widths*: the closed formula $w(\sigma) = N / \gcd(N, c^2)$, its minimality, invariance under the full representative change $\sigma \mapsto \gamma\sigma T^k$ ($\gamma \in \Gamma_0(N)$, $k \in \mathbb{Z}$), the equivalence

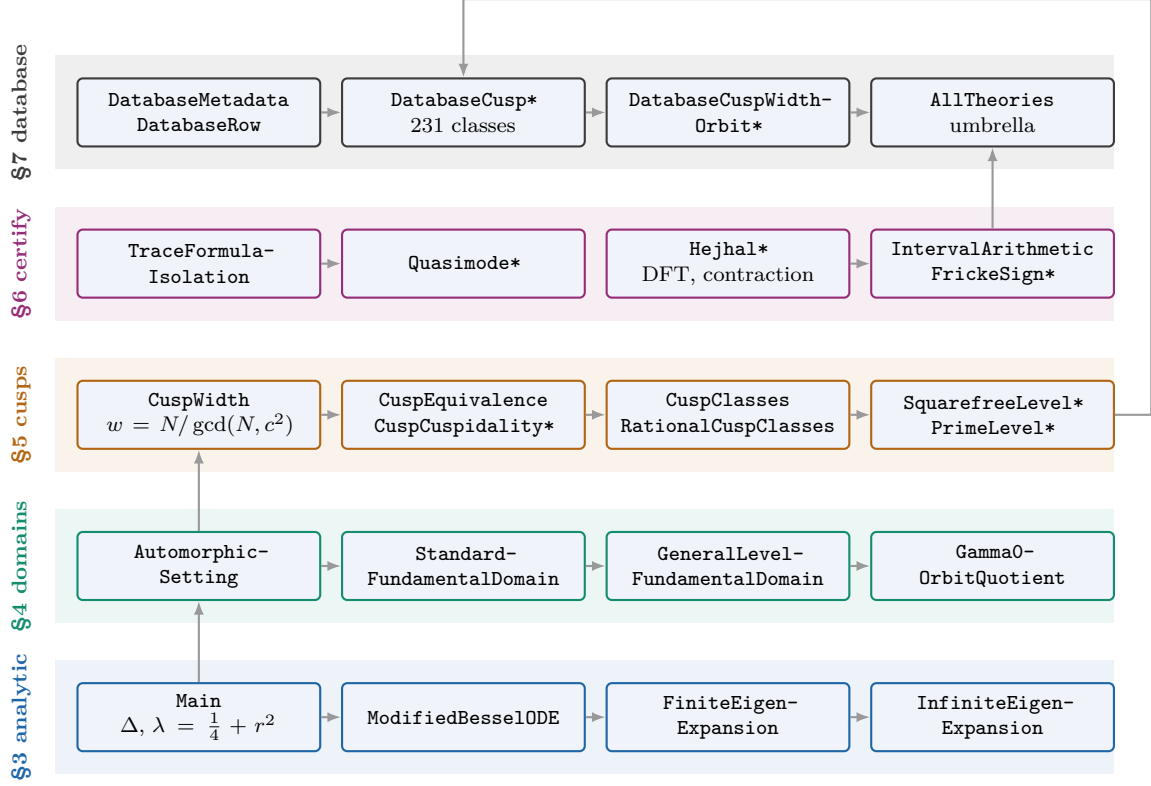


Figure 1: Architecture of the development. Arrows denote genuine module dependencies; each band corresponds to one section of this paper. Names in `typewriter` are Lean modules in `RequestProject/`.

of the “all certified periods” and “minimal width” formulations of cuspidality, the divisibility classification, the exact realization theorems at prime and squarefree level, and a canonical equivalence with the geometric model $\mathbb{P}^1(\mathbb{Q})/\Gamma_0(N)$ (Section 5).

4. Finite and abstract certification cores corresponding to the three algorithms, with sound center-error interval propagation and a contraction-based Hejhal stopping criterion (Section 6).
5. A complete formal accounting of the announced snapshot’s cusp combinatorics, including the 231 level-tagged classes, the width distribution and its exact support, the complementary-width involution, its 116 orbits, and the structural decomposition $231 = 1 + 2 \cdot 115$ (Section 7).
6. An explicit, itemized faithfulness audit separating transcription, proof, implementation and empirical validation (Section 9).

2 The formal framework

2.1 Logic, kernel and trusted base

The development is written in Lean 4 (version 4.28.0) on top of Mathlib (commit `8f9d9cff6bd7`). Statements live in dependent type theory; a completed proof is a term whose type is the statement, re-checked by the Lean kernel. The library declares *no* axioms of its own. Its results depend only on the standard Lean axioms `propext`, `Quot.sound` and `Classical.choice`, except for those closed by native evaluation.

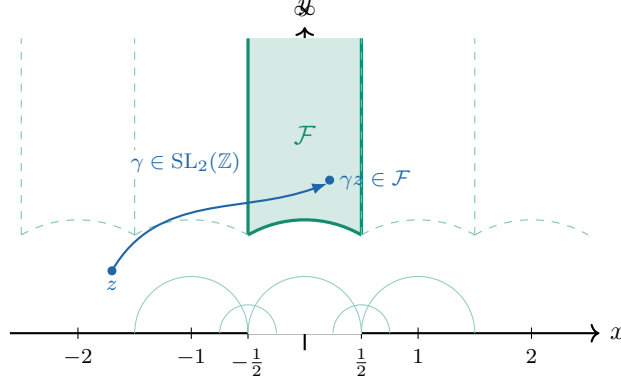


Figure 2: The upper half-plane, the standard modular fundamental domain $\mathcal{F} = \{|x| \leq \frac{1}{2}, x^2 + y^2 \geq 1\}$ and a pullback. Formally: `MaassAutomorphic.standardFDCoordinates` is the displayed (x, y) region and `MaassAutomorphic.coordinates_mem_standardFDCoordinates_iff` identifies it with Mathlib’s `ModularGroup.fd`; the existence of the arrow is `MaassAutomorphic.pullbackPoint_mem_fd`.

Remark 2.1 (Native decision and the trusted base). 25 of the 82 modules use the tactic `native_decide`, which discharges a decidable proposition by compiling it and running it, and hence depends on the additional axioms `Lean.ofReduceBool` and `Lean.trustCompiler`. This is how finite verifications such as “there are exactly 65 squarefree levels ≤ 105 ” or “the sum of the 231 exact widths is 4640” are closed. These results are therefore checked by the Lean *compiler* rather than by the kernel alone; the distinction is recorded wherever such a theorem is quoted below with the tag (*).

2.2 How to read the statements

Every displayed theorem carries the name of the Lean declaration proving it and the module containing it, in the form `Namespace.declaration [Module]`. Where the Lean statement is literally the displayed statement, we say so; where the rendering elides technical hypotheses (typically differentiability or positivity side conditions), those hypotheses are described in words.

3 The analytic core

3.1 The upper half-plane and its action

Let $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{R})$ the Möbius action is $\gamma z = \frac{az+b}{cz+d}$.

Theorem 3.1 (Imaginary part transformation). *For every $\gamma \in \text{SL}_2(\mathbb{R})$ and $z \in \mathbb{H}$,*

$$\text{Im}(\gamma z) = \frac{\text{Im } z}{|cz + d|^2}.$$

`MaassForms.im_smul_eq_div_normSq_SL [Main.lean]`

Because the right-hand side is positive, this is exactly the statement that the action preserves $\mathbb{H}$. Figure 2 illustrates the action and the standard fundamental domain.

3.2 The Laplacian and the spectral parameter

In coordinates $z = x + iy$ the hyperbolic Laplace–Beltrami operator is $\Delta = y^2(\partial_x^2 + \partial_y^2)$, formalized as `MaassForms.laplaceBeltrami`, and a *Maass-type eigenfunction* with eigenvalue λ is a function satisfying $\Delta f + \lambda f = 0$ at every point of $\mathbb{H}$ (`MaassForms.isLaplaceEigenform`).

Theorem 3.2 (Constant mode and spectral relation). *Write $y^s = \exp(s \log y)$ for $y > 0$. Then*

$$\Delta(y^s) = s(s-1)y^s, \quad -\left(\frac{1}{2} + ir\right)\left(\frac{1}{2} + ir - 1\right) = \frac{1}{4} + r^2,$$

and consequently $y^{1/2+ir}$ is a Δ -eigenfunction with eigenvalue $\lambda = \frac{1}{4} + r^2$.

`MaassForms.laplaceBeltrami_ypow`, `MaassForms.spectral_parameter`,
`MaassForms.ypow_isLaplaceEigenform` [Main.lean]

3.3 Separation of variables and the Bessel equation

The Fourier expansion of a Maass cusp form,

$$f(x + iy) = \sqrt{y} \sum_{n \neq 0} a_n K_{ir}(2\pi|n|y) e^{2\pi i n x},$$

rests on the fact that a single mode $f(x, y) = e^{2\pi i n x} W(y)$ satisfies the eigenvalue equation exactly when W satisfies a second-order ODE. This is proved in full.

Theorem 3.3 (Separation of variables). *Let $n \in \mathbb{Z}$, $\lambda \in \mathbb{C}$, and let W be twice differentiable. Then, for all x and all $y > 0$,*

$$\Delta f + \lambda f = e^{2\pi i n x} \left(y^2 W''(y) - 4\pi^2 n^2 y^2 W(y) + \lambda W(y) \right), \quad f(x, y) = e^{2\pi i n x} W(y).$$

`MaassForms.separation_of_variables` [Main.lean]

Theorem 3.4 (From the Bessel equation to an eigenfunction). *Say that (K, K_1, K_2) solves the modified Bessel equation of order r if K_1, K_2 are successive derivatives of K and*

$$t^2 K_2(t) + t K_1(t) - (t^2 - r^2) K(t) = 0 \quad \text{for all } t.$$

Then for every $n \in \mathbb{Z}$ the function $f(x, y) = e^{2\pi i n x} \sqrt{y} K(2\pi|n|y)$ is a Δ -eigenfunction with eigenvalue $\lambda = \frac{1}{4} + r^2$.

`MaassForms.modifiedBessel_fourierMode_isLaplaceEigenform` [ModifiedBesselODE.lean]

Theorem 3.4 is the substantive local content of expansion (1) of the announcement that can be formalized without an analytic theory of K_{ir} : the rescaling $y \mapsto \sqrt{y} K(ay)$ is proved to convert the Bessel equation into the radial Maass equation. What is *not* available is the construction of the distinguished decaying solution K_{ir} itself (Section 9).

3.4 Superposition

Two further modules make the passage from single modes to expansions precise. `FiniteEigenExpansion` proves that the coordinate Laplacian is linear under explicit smoothness hypotheses, so a finite linear combination of modes with a common eigenvalue is again an eigenfunction. `InfiniteEigenExpansion` isolates the analytic hypotheses required for an infinite sum—summability of the values and of the termwise first and second derivatives, together with the validity of termwise differentiation, packaged as the structure `MaassForms.HasTermwiseLaplaceExpansion`—and proves:

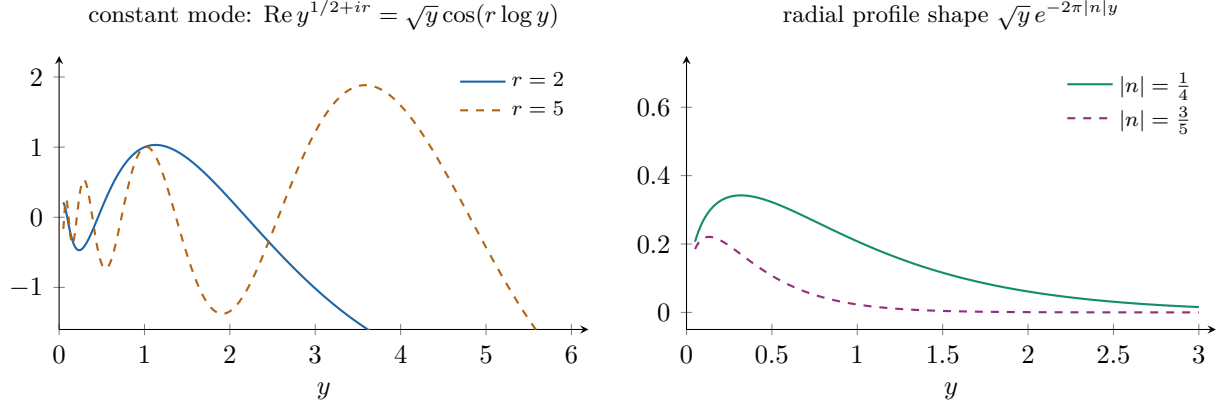


Figure 3: Left: the real part of the formalized constant mode $y^{1/2+ir}$ (Theorem 3.2). Right: the shape $\sqrt{y} e^{-2\pi|n|y}$ of a decaying radial profile; the exponential stands in for K_{ir} , whose analytic theory is *not* part of the development. Only the left curve corresponds to a formalized function; the right is illustrative.

Theorem 3.5 (Termwise expansion). *If $(c_i)_{i \in \iota}$ and (f_i) satisfy MaassForms.HasTermwiseLaplaceExpansion and each f_i is a Δ -eigenfunction with the same eigenvalue λ , then $\sum_i c_i f_i$ is a Δ -eigenfunction with eigenvalue λ .*

MaassForms.tsum_isLaplaceEigenform [InfiniteEigenExpansion.lean]

This is an honest conditional statement: the hypotheses are assumed, not established for the Bessel series.

4 Automorphy, fundamental domains and cuspidality

4.1 The $\Gamma_0(N)$ package

Let $\Gamma_0(N) \leq \mathrm{SL}_2(\mathbb{Z})$ be the usual congruence subgroup (Mathlib's `CongruenceSubgroup.Gamma0`). A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is $\Gamma_0(N)$ -*automorphic of weight zero* if $f(\gamma z) = f(z)$ for every $\gamma \in \Gamma_0(N)$ (`MaassAutomorphic.IsGamma0Automorphic`). The development proves the expected closure properties (zero, sums, scalar multiples), that the translation $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ lies in $\Gamma_0(N)$ for every N , and hence that every automorphic function is 1-periodic in x (`MaassAutomorphic.IsGamma0Automorphic.one_periodic`).

Cuspidality at infinity is the vanishing of the constant Fourier coefficient at every height:

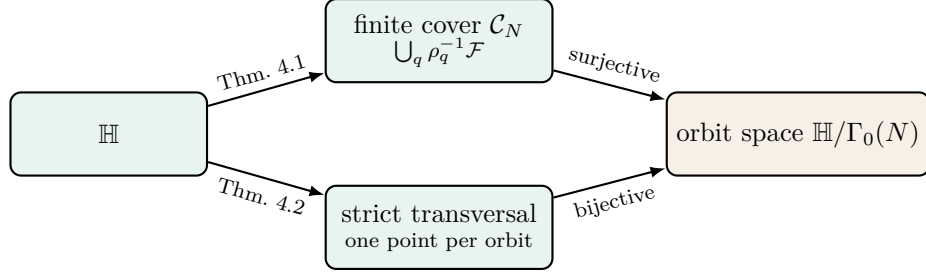
$$\int_0^1 f(x + iy) \, dx = 0 \quad \text{for all } y > 0,$$

together with interval integrability of the integrand (`MaassAutomorphic.IsCuspidalAtInfinity`). Square-integrability is formalized with respect to the hyperbolic measure on a chosen coordinate region (`MaassAutomorphic.IsHyperbolicSquareIntegrableOn`), and the three conditions are bundled into the structure `MaassAutomorphic.IsGamma0MaassFormCuspidalAtInfinity`.

4.2 Fundamental domains at level one and at general level

At level one the development identifies the explicit region

$$\mathcal{F} = \{(x, y) : y > 0, x^2 + y^2 \geq 1, |x| \leq \tfrac{1}{2}\}$$



 formalized: existence, finiteness, exactness of the transversal.
 not formalized: measurable structure, bound-
 ary identifications, a strict geometric region.

Figure 4: The two formal substitutes for a fundamental domain at level N . The finite cover is geometric but not strict; the transversal is strict but noncomputable and carries no measurable structure.

with Mathlib’s standard modular domain `ModularGroup.fd`, and constructs a pullback: every $z \in \mathbb{H}$ admits $\gamma \in \text{SL}_2(\mathbb{Z})$ with $\gamma z \in \mathcal{F}$ (`MaassAutomorphic.pullbackPoint_mem_fd`); the choice is noncomputable, made with `Classical.choice`.

At general level, finiteness of the index $[\text{SL}_2(\mathbb{Z}) : \Gamma_0(N)]$ gives a finite cover.

Theorem 4.1 (Finite fundamental cover). *For every $N \geq 1$ let $\mathcal{C}_N = \{z : \exists q, \rho_q z \in \mathcal{F}\}$, where ρ_q runs over a set of representatives of the right cosets of $\Gamma_0(N)$. Then $\mathcal{C}_N$ is a finite union of preimages of $\mathcal{F}$, and every $z \in \mathbb{H}$ has a $\Gamma_0(N)$ -translate in $\mathcal{C}_N$.*

`MaassAutomorphic.gamma0FundamentalCover_eq_finset_iUnion,`
`MaassAutomorphic.exists_gamma0_smul_mem_fundamentalCover [GeneralLevelFundamentalDomain.lean]`

Theorem 4.2 (Strict orbit transversal). *The quotient $\mathbb{H}/\Gamma_0(N)$ exists as a setoid quotient, and the set of chosen orbit representatives is a strict transversal: every orbit meets it, and two representatives that are $\Gamma_0(N)$ -equivalent are equal.*

`MaassAutomorphic.exists_gamma0_equiv_strictTransversal,`
`MaassAutomorphic.gamma0StrictTransversal_eq_of_rel [Gamma0OrbitQuotient.lean]`

5 Cusps and exact widths

This section describes the mathematically deepest part of the development: roughly twenty modules develop the theory of cusps of $\Gamma_0(N)$, from the width formula to a complete classification at squarefree level.

5.1 Scaling matrices and the exact width

A cusp of $\Gamma_0(N)$ is represented by a *scaling matrix* $\sigma \in \text{SL}_2(\mathbb{Z})$, thought of as sending ∞ to the cusp $\sigma\infty \in \mathbb{P}^1(\mathbb{Q})$. The relevant invariant is the smallest $w > 0$ with $\sigma T^w \sigma^{-1} \in \Gamma_0(N)$.

Definition 5.1 (Exact width). For $N \geq 1$ and $\sigma \in \text{SL}_2(\mathbb{Z})$ with lower-left entry $c = \sigma_{10}$, put

$$w_N(\sigma) := \frac{N}{\gcd(N, |c|^2)}.$$

`MaassAutomorphic.cuspWidth [CuspWidth.lean]`

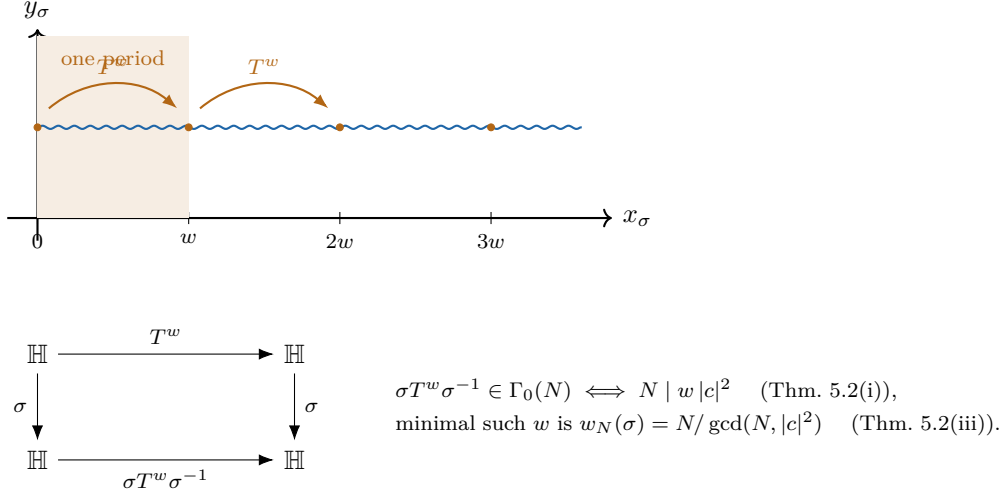


Figure 5: Local coordinates at a cusp. In the coordinate $z \mapsto \sigma^{-1}z$ the stabilizer acts by the translation T^w , and the exact width $w = w_N(\sigma)$ is the length of one period. The commuting square is the content of the conjugation criterion.

Theorem 5.2 (Characterization, membership and minimality). *For all $N, w \geq 0$ and $\sigma \in \mathrm{SL}_2(\mathbb{Z})$:*

- (i) $\sigma T^w \sigma^{-1} \in \Gamma_0(N) \iff N \mid w|c|^2$;
- (ii) if $N > 0$ then $w_N(\sigma) > 0$ and $\sigma T^{w_N(\sigma)} \sigma^{-1} \in \Gamma_0(N)$;
- (iii) if $N > 0$ and $\sigma T^w \sigma^{-1} \in \Gamma_0(N)$ then $w_N(\sigma) \mid w$; in particular $w_N(\sigma)$ is minimal among positive stabilizing exponents.

`MaassAutomorphic.conjugate_T_pow_mem_Gamma0_iff`, `MaassAutomorphic.cuspWidth_pos`,
`MaassAutomorphic.conjugate_T_pow_cuspWidth_mem_Gamma0`,
`MaassAutomorphic.cuspWidth_dvd_of_conjugate_T_pow_mem_Gamma0`

5.2 Independence of the representative

A cusp is not a matrix: two scaling matrices represent the same cusp when they differ by $\sigma \mapsto \gamma \sigma T^k$ with $\gamma \in \Gamma_0(N)$, $k \in \mathbb{Z}$. The development proves that everything of interest is invariant under this change.

Theorem 5.3 (Invariance). *Let $N > 0$, $\gamma \in \Gamma_0(N)$, $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, $k \in \mathbb{Z}$. Then*

$$w_N(\gamma \sigma T^k) = w_N(\sigma),$$

and for every $\Gamma_0(N)$ -automorphic f the local cusp expression at $\gamma \sigma T^k$ is obtained from that at σ by translating the cusp coordinate. Consequently interval integrability over one exact period and the vanishing of the constant term are preserved.

`MaassAutomorphic.cuspWidth_equivalent_rep`, `MaassAutomorphic.at_minimalCusp_equivalent_rep`
`[CuspEquivalence.lean]`, `CuspCuspidalityEquivalence.lean`, `CuspIntegerTranslation.lean`

Theorem 5.4 (Minimal-width cuspidality). *For a $\Gamma_0(N)$ -automorphic function at positive level, the condition “the constant term vanishes over every certified stabilizing period at every integral representative” is equivalent to the condition “the constant term vanishes over the exact minimal width at every integral representative”.*

`MaassAutomorphic.isCuspidalAtAllCusps_iff_minimalRepresentatives` `[CuspCuspidalityMinimal.lean]`

Theorem 5.5 (Level one). *For a $\mathrm{SL}_2(\mathbb{Z})$ -automorphic f , cuspidality at all cusps is equivalent to cuspidality at infinity; hence the standard-domain level-one Maass cusp form package is equivalent to its strengthened all-cusps version.*

MaassAutomorphic.isCuspidalAtAllCusps_one_iff_atInfinity,
MaassAutomorphic.isLevelOneMaassCuspFormAllCusps_iff [LevelOneCuspidality.lean]

5.3 Classification of widths

Theorem 5.6 (Divisibility and extremes). *For every N and σ : $w_N(\sigma) \mid N$; hence the set of realized widths is finite. For $N > 0$, $w_N(\sigma) = 1$ iff $N \mid |c|^2$, and $w_N(\sigma) = N$ iff $\gcd(N, |c|^2) = 1$.*

MaassAutomorphic.cuspWidth_dvd_level, MaassAutomorphic.cuspWidth_eq_one_iff,
MaassAutomorphic.cuspWidth_eq_level_iff, MaassAutomorphic.finite_setOf_cuspWidth
[CuspWidthClassification.lean]

Theorem 5.7 (Prime and squarefree level). *Let p be prime. The realized widths at level p are exactly 1 and p , classified by whether $p \mid c$, and both are realized (by ∞ and 0). For every positive squarefree N , the realized exact widths are precisely the divisors of N , each realized by an explicit lower-unipotent representative.*

[PrimeLevelCuspWidths.lean, SquarefreeLevelCuspWidths.lean]

5.4 Cusp classes: algebraic and geometric models

Two models of the set of cusps are constructed and compared.

Definition 5.8. $\mathrm{CuspClass}(N)$ is the quotient of $\mathrm{SL}_2(\mathbb{Z})$ by the relation $\sigma \sim \gamma\sigma T^k$; $\mathrm{RationalCuspClass}(N)$ is the quotient of $\mathbb{P}^1(\mathbb{Q}) = \mathrm{OnePoint} \ \mathbb{Q}$ by the $\Gamma_0(N)$ -action.
[CuspClasses.lean, RationalCuspClasses.lean]

Theorem 5.9 (Comparison and finiteness). *For every $N > 0$ the two models are canonically equivalent, both are finite, and the exact width descends to a well-defined function on either. Moreover every rational projective cusp has an integral scaling representative.*

MaassAutomorphic.cuspClassEquivRational and the finiteness instances

Theorem 5.10 (Complete classification). *Let $N > 0$.*

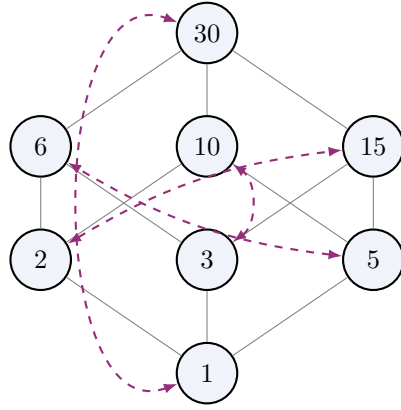
- (i) *If $N = p$ is prime, every cusp is equivalent to ∞ or 0, these are distinct, and both models have cardinality 2.*
- (ii) *If N is squarefree, the exact width is a complete invariant: two classes coincide iff their widths agree; each model is equivalent to the set of positive divisors of N ; hence*

$$\# \mathrm{CuspClass}(N) = \# \mathrm{RationalCuspClass}(N) = \tau(N) = 2^{\omega(N)},$$

where $\omega(N)$ is the number of prime divisors.

MaassAutomorphic.card_cuspClass_prime, MaassAutomorphic.cuspClass_eq_iff_width_eq_of_squarefree,
MaassAutomorphic.card_rationalCuspClass_squarefree_eq_pow_primeFactors

Figure 6 shows the resulting picture at $N = 30$, together with the complementary-width duality used in Section 7.



The involution $w \mapsto N/w$ pairs the 8 classes into 4 orbits.

$$w \longleftrightarrow N/w$$

$N = 30 = 2 \cdot 3 \cdot 5$ is squarefree, so the $2^{\omega(30)} = 8$ cusp classes of $\Gamma_0(30)$ are labelled bijectively by the divisors, the label being the exact width (Theorem 5.10(ii)).

Figure 6: Cusps of $\Gamma_0(30)$: the divisor lattice with exact widths as labels, and the complementary-width involution.

6 Certification cores

The announcement's forms are certified by three methods. The development formalizes finite or abstract cores of each, plus the interval calculus that all three need. We are explicit: these are faithful models of the *mathematical mechanism*, not implementations of the algorithms (Section 9).

6.1 Center-error intervals

Definition 6.1. A *center-error pair* over a pseudo-metric space α consists of a center $c \in \alpha$ and an error $\varepsilon \in \mathbb{R}_{\geq 0}$; it *certifies* $x \in \alpha$ when $d(c, x) \leq \varepsilon$.

`HejhalHorocycle.CenterError, HejhalHorocycle.CenterError.Certifies`

Theorem 6.2 (Sound propagation). *Over $\mathbb{R}$, if a certifies x and b certifies y , then $a \oplus b$ certifies $x + y$, $\ominus a$ certifies $-x$, $a \ominus b$ certifies $x - y$, and*

$$a \otimes b := (c_a c_b, |c_a| \varepsilon_b + |c_b| \varepsilon_a + \varepsilon_a \varepsilon_b)$$

certifies xy . Moreover, for a K -Lipschitz f , the image pair $(f(c), K\varepsilon)$ certifies $f(x)$, and this map composes with constant KL .

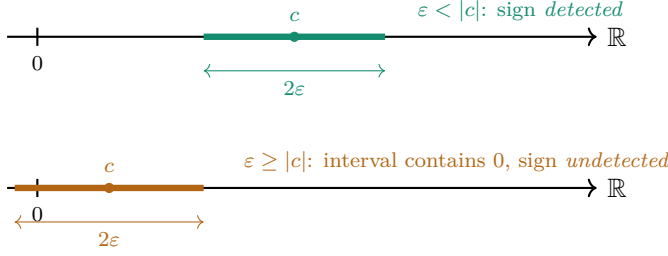
`[IntervalArithmetic.lean, CertifiedComputation.lean]`

Theorem 6.3 (Fricke-sign detection). *Call a pair definitely nonzero when $\varepsilon < |c|$. Then a pair is definitely nonzero iff it is definitely positive or definitely negative, iff its interval excludes 0; a definitely nonzero pair certifying x forces $x \neq 0$; and from the sign relation $a_- = w a_+$ the Fricke sign w is determined by the signs of any certified nonzero pair (a_+, a_-) .*

`[FrickeSignCertification.lean]`

6.2 Trace-formula isolation

The rigorous trace formula is used to isolate spectral data by a weighted sum over the spectrum. The formalized core keeps the weighting and drops the geometric side.



In the announced snapshot, 15,423 of the 35,416 rows fall in the second case, leaving 19,993 rows with a detected Fricke sign (`MaassDatabase.announced_fricke_partition`).

Figure 7: Sign detection from a center-error interval (Theorem 6.3), and the corresponding split of the announced snapshot.

Theorem 6.4 (Isolation). *Let S be a finite index set, $F : \mathbb{R} \rightarrow \mathbb{R}$ a test function, and set $\Sigma = \sum_{j \in S} F(r_j) a_j$. If $F(r_{j_0}) = 1$ and $F(r_j) = 0$ for $j \neq j_0$, then $\Sigma = a_{j_0}$. In general*

$$\left| \Sigma - F(r_{j_0}) a_{j_0} \right| \leq \sum_{j \in S \setminus \{j_0\}} |F(r_j) a_j| \leq A \sum_{j \in S \setminus \{j_0\}} |F(r_j)| \quad \text{when } |a_j| \leq A.$$

`MaassTraceFormula.exact_isolation`, `MaassTraceFormula.approximate_isolation`,
`MaassTraceFormula.approximate_isolation_of_coefficient_bound` [`TraceFormulaIsolation.lean`]

Theorem 6.5 (Window counting and emptiness). *With nonnegative weights that are $\geq \eta$ on a window $[lo, hi]$,*

$$\eta \cdot \#\{j \in S : r_j \in [lo, hi]\} \leq \sum_{j \in S} \text{weight}(j),$$

and if all weights on the window are strictly positive while the total weight vanishes, then no r_j lies in the window.

`MaassTraceFormula.window_count_bound`, `MaassTraceFormula.no_spectrum_in_window`

6.3 Quasimode certification

Theorem 6.6 (Small residual forces proximity). *Let S be finite, $(\lambda_i)_{i \in S}$ eigenvalues, (c_i) coefficients with $\sum_i c_i^2 = 1$, and let $R(\mu) = \sum_i ((\lambda_i - \mu) c_i)^2$. If $\varepsilon \geq 0$ and $R(\mu) < \varepsilon^2$, then $|\lambda_i - \mu| < \varepsilon$ for some $i \in S$. More precisely $\varepsilon^2 \sum_i c_i^2 \leq R(\mu)$ whenever all $|\lambda_i - \mu| \geq \varepsilon$.*

`MaassQuasimode.quasimode_near_spectrum`, `MaassQuasimode.residualEnergy_lower_bound`
`[QuasimodeCertification.lean]`

Theorem 6.7 (Concentration). *With the same notation, the near and far windows partition S , and*

$$\sum_{|\lambda_i - \mu| \geq \varepsilon} c_i^2 \leq \frac{R(\mu)}{\varepsilon^2}, \quad R(\mu) \leq \delta \varepsilon^2 \implies \sum_{|\lambda_i - \mu| < \varepsilon} c_i^2 \geq 1 - \delta.$$

`MaassQuasimode.far_coefficientEnergy_le`, `MaassQuasimode.near_coefficientEnergy_ge`
`[QuasimodeConcentration.lean]`

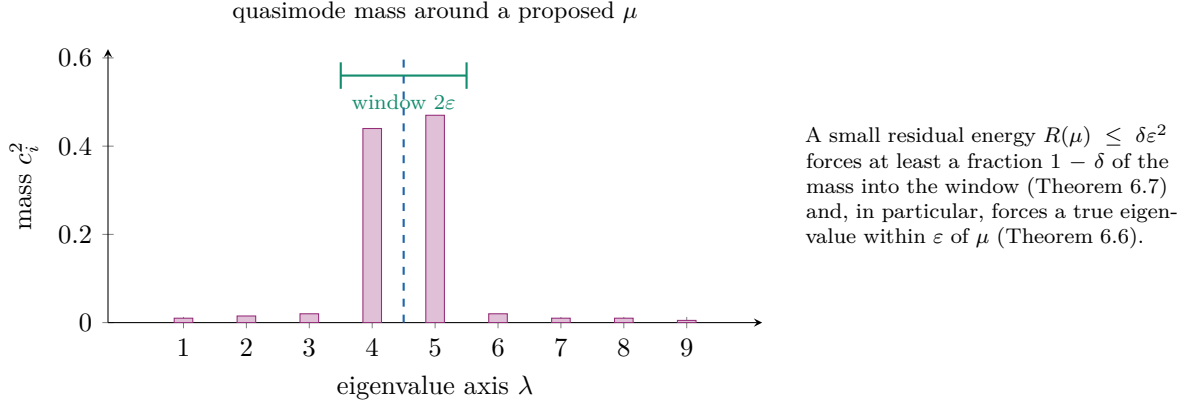


Figure 8: The formalized quasimode mechanism, in a finite spectral model.

6.4 Hejhal’s algorithm: horocycle extraction

Hejhal’s method samples an automorphic function along a horocycle $y = Y$ at the points $x_j = j/N$, $0 \leq j < N$, and recovers Fourier coefficients by a discrete Fourier transform. Write $\omega_N(t) = e^{2\pi it/N}$.

Theorem 6.8 (Orthogonality). *For $N > 0$ and $k \in \mathbb{Z}$,*

$$\sum_{j=0}^{N-1} \omega_N(kj) = \begin{cases} N, & N \mid k, \\ 0, & \text{otherwise.} \end{cases}$$

HejhalHorocycle.dft_orthogonality [HejhalHorocycle.lean]

Theorem 6.9 (Master horocycle identity and aliasing). *Let $a : \mathbb{Z} \rightarrow \mathbb{C}$ be supported on a finite set S , let $v(j) = \sum_{m \in S} a_m \omega_N(mj)$ be the sampled signal and let*

$$\widehat{v}(n) = \frac{1}{N} \sum_{j=0}^{N-1} v(j) \omega_N(-nj)$$

be the horocycle average. Then

$$\widehat{v}(n) = \sum_{m \in S, N \mid m-n} a_m.$$

Consequently $\widehat{v}(n) = a_n$ whenever no other frequency of S is congruent to n modulo N , and in general

$$|\widehat{v}(n) - a_n| \leq \sum_{m \in S \setminus \{n\}, N \mid m-n} |a_m|.$$

HejhalHorocycle.horocycle_master, HejhalHorocycle.horocycle_extract_exact,
HejhalHorocycle.extract_exact_of_no_alias, HejhalHorocycle.horocycle_aliasing_bound
[HejhalHorocycle.lean, HejhalIteration.lean]

At level one the sampling is compatible with pullback into the fundamental domain: each sample may be replaced by its pullback without changing the weighted average.

Theorem 6.10 (Pullback invariance). *If f is $\mathrm{SL}_2(\mathbb{Z})$ -automorphic then, for every $N, Y > 0$ and $n \in \mathbb{Z}$, the pulled-back horocycle average equals the plain horocycle average, and every pulled-back sample lies in $\mathcal{F}$.*

HejhalPullback.pulledbackAverage_eq_automorphicAverage,
HejhalPullback.pulledback_horocyclePoint_mem_fd [HejhalPullback.lean]

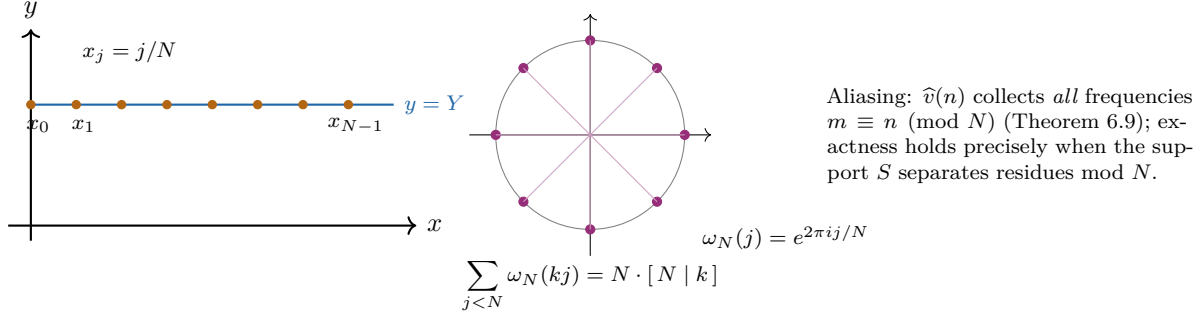


Figure 9: Horocycle sampling and the orthogonality that makes coefficient extraction exact.

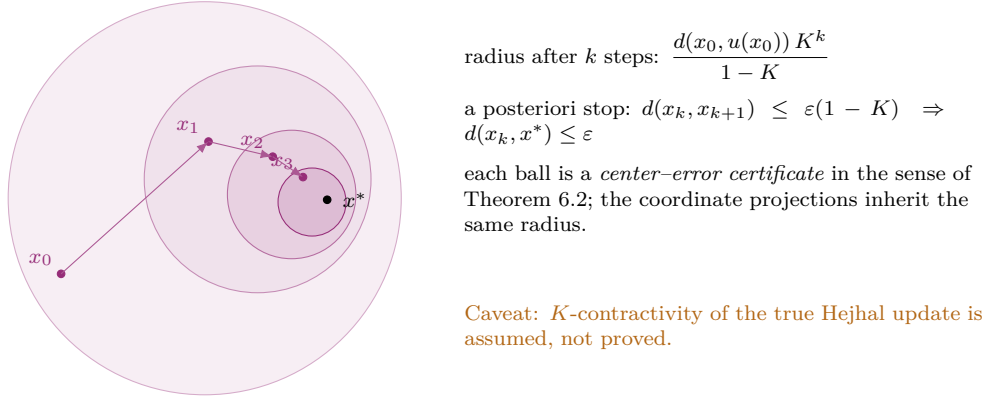


Figure 10: The formalized Hejhal iteration: geometric convergence with a certified radius (Theorem 6.11).

6.5 Hejhal's algorithm: iteration and certificates

The Hejhal update is modelled abstractly as a self-map of a state space $\text{HejhalState}(Q) = \mathbb{R} \times (\text{Fin } Q \rightarrow \mathbb{C})$ (a spectral parameter together with Q coefficients). Contraction is a *hypothesis*: the development does not prove that the genuine update contracts.

Theorem 6.11 (Convergence and a posteriori certificate). *Let u be a K -contraction of $\text{HejhalState}(Q)$ with fixed point x^* , and let $x_k = u^k(x_0)$. Then $x_k \rightarrow x^*$,*

$$d(x_k, x^*) \leq \frac{d(x_0, u(x_0)) K^k}{1 - K},$$

the pair $(x_k, \text{that bound})$ certifies x^ , both the height coordinate and each coefficient converge with the same bound, and the stopping criterion*

$$d(x_k, x_{k+1}) \leq \epsilon(1 - K) \implies d(x_k, x^*) \leq \epsilon$$

holds. Finally, an error δ in the spectral parameter propagates to the eigenvalue as $|(\frac{1}{4} + r^2) - (\frac{1}{4} + r_0^2)| \leq \delta(2|r_0| + \delta)$.

HejhalHorocycle.hejhal_iteration_converges, HejhalHorocycle.hejhal_iteration_error_le,
HejhalHorocycle.hejhalCertificate_certifies, HejhalHorocycle.hejhal_stop_when_residual_small,
HejhalHorocycle.spectral_parameter_center_error [HejhalIteration.lean, HejhalCertification.lean]

7 The announced snapshot as formal arithmetic

7.1 Metadata

The constants of the announcement are recorded in a structure `MaassDatabase.SnapshotMetadata` and instantiated as `MaassDatabase.announcedSnapshot`: 35,416 forms; 2,202 of level one; 1,000 coefficients per row; 15,423 rows with undetected Fricke sign; 4954 MB of storage. From these, arithmetic consequences are proved:

Theorem 7.1 (Metadata arithmetic). $35,416 = 2,202 + 33,214$; $35,416 = 15,423 + 19,993$; *the number of stored coefficient intervals is $35,416 \times 1,000 = 35,416,000$, and with one eigenvalue interval per row, 35,451,416 scalar intervals; the storage is $4954/1000 = 2477/500$ GB.*

`MaassDatabase.announced_level_split, MaassDatabase.announced_fricke_partition,`
`MaassDatabase.announced_coefficient_interval_count, MaassDatabase.announced_scalar_interval_count,`
`MaassDatabase.announced_storage_as_gigabytes [DatabaseMetadata.lean]`

Theorem 7.2 (Supported levels, $(*)$). $\text{supportedLevels} = \{N : 1 \leq N \leq 105, N \text{ squarefree}\}$ *has exactly 65 elements, contains 1 and 105, and consists of positive integers.*

`MaassDatabase.supportedLevels_card, MaassDatabase.mem_supportedLevels_iff`

Remark 7.3. Theorem 7.1 is arithmetic about the announcement's numbers. It does not validate them: the snapshot itself is not present in the repository.

7.2 Cusp accounting

Combining Theorem 5.10 with Theorem 7.2 yields the exact cusp combinatorics of the announced level set.

Theorem 7.4 (231 level-tagged cusp classes, $(*)$).

$$\sum_{N \in \text{supportedLevels}} \# \text{RationalCuspClass}(N) = \sum_{N \in \text{supportedLevels}} \tau(N) = 231,$$

and the dependent-sum type `SupportedRationalCuspClass` of all level-tagged rational cusp classes has cardinality 231.

`MaassDatabase.supportedCuspClassCount_eq, MaassDatabase.sum_card_rationalCuspClass_supportedLevels`
`[DatabaseCuspAccounting.lean, DatabaseCuspDistribution.lean]`

Theorem 7.5 (Level histogram, $(*)$). *Every supported level has 1, 2, 4 or 8 cusp classes, with respective multiplicities 1, 27, 30, 7; and*

$$1 + 27 + 30 + 7 = 65, \quad 1 \cdot 1 + 27 \cdot 2 + 30 \cdot 4 + 7 \cdot 8 = 231.$$

`MaassDatabase.supportedLevel_cuspCount_cases, MaassDatabase.supported_cusp_histogram_level_total,`
`MaassDatabase.supported_cusp_histogram_weighted_total`

Theorem 7.6 (Width distribution, $(*)$). *Define the multiplicity of a width w as the number of supported levels having w as a divisor. Then*

$$m(1) = 65, \quad m(2) = 21, \quad m(3) = 17, \quad m(5) = 11, \quad m(7) = 9,$$

$m(w) = 0$ for $w > 105$, $\sum_{w=1}^{105} m(w) = 231$, and

$$\sum_{w=1}^{105} w m(w) = \sum_{N \in \text{supportedLevels}} \sum_{d|N} d = 4640.$$

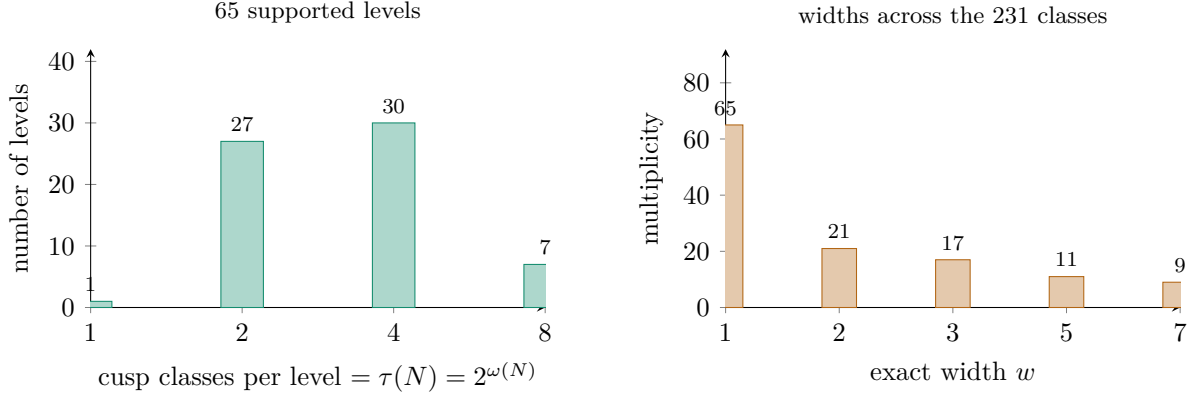


Figure 11: Left: Theorem 7.5. Right: the first five multiplicities of the width distribution, each an individually formalized value, (*) (Theorem 7.6); widths $w > 7$ occur with smaller multiplicities and are not displayed.

Moreover the divisor coordinate of a level-tagged class coincides with its geometric exact width.

[DatabaseCuspWidthStatistics.lean]

Theorem 7.7 (Exact support, (*)). *The set of widths realized across all supported levels is exactly the set of supported levels: a positive integer occurs as a geometric exact width of some level-tagged class iff it is squarefree and at most 105. Equivalently $m(w) > 0$ iff $1 \leq w \leq 105$ and w is squarefree.*

MaassDatabase.supportedCuspWidths_eq_supportedLevels, MaassDatabase.exists_geometric_cusp_width_iff
[DatabaseCuspWidthSupport.lean]

7.3 Complementary-width duality and the orbit decomposition

The involution $w \mapsto N/w$ on divisors transports, through the equivalence of Theorem 5.10, to a canonical involution on the 231 level-tagged classes.

Theorem 7.8 (Duality). *There is an involution $c \mapsto c^\vee$ on SupportedRationalCuspClass preserving the level, with*

$$w(c^\vee) = \frac{N}{w(c)}, \quad w(c) \cdot w(c^\vee) = N,$$

and $c^\vee = c$ iff the level of c is 1; in particular there is exactly one fixed point.

[DatabaseCuspWidthDuality.lean]

Theorem 7.9 (Orbits, (*)). *Selecting in each pair the class of smaller width gives 116 canonical representatives. The genuine setoid quotient by the duality relation is equivalent to this subtype, so the orbit space has cardinality 116; it consists of exactly one fixed orbit and 115 nontrivial orbits. On the level of fibers, the fixed orbit contains one supported class and every nontrivial orbit contains exactly two.*

MaassDatabase.card_supportedCuspDualRepresentative,
MaassDatabase.supportedCuspDualOrbitToRepresentative_bijective,
MaassDatabase.card_fixedSupportedCuspDualOrbit, MaassDatabase.card_nontrivialSupportedCuspDualOrbit
[DatabaseCuspWidthOrbits.lean, DatabaseCuspWidthOrbitQuotient.lean,
DatabaseCuspWidthOrbitStructure.lean]

Theorem 7.10 (Structural decomposition). *There is an explicit bijection*

$$(\text{fixed orbits}) \sqcup (\text{nontrivial orbits} \times \{0, 1\}) \xrightarrow{\sim} \text{SupportedRationalCuspClass},$$

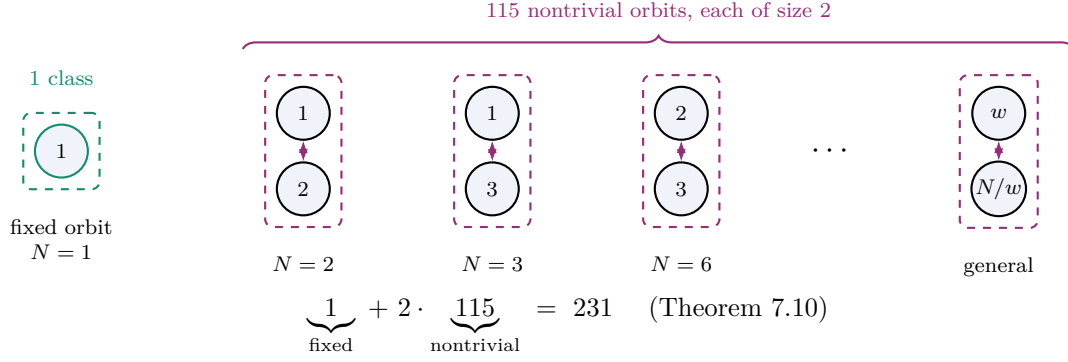


Figure 12: The complementary-width duality on the 231 level-tagged cusp classes of the announced level set: one fixed orbit and 115 two-element orbits.

whence

$$231 = \#\{\text{fixed}\} + 2 \#\{\text{nontrivial}\} = 1 + 2 \cdot 115.$$

MaassDatabase.supportedCuspOrbitDecomposition, MaassDatabase.supportedCuspOrbit_decomposition_count, MaassDatabase.supportedCuspOrbit_decomposition_count_numeric [DatabaseCuspWidthOrbitDecomposition.lean]

7.4 Rows and certificates

DatabaseRow and SpectralCertification give generic schemas for a database row (a spectral parameter interval together with a finite list of coefficient intervals) and for what it means for such a row to be certified, built on the center–error calculus of Theorem 6.2. They are schemas: no announced row is instantiated in the repository.

8 Beyond the announcement

A substantial part of the repository proves verified mathematics that is *adjacent* to, rather than part of, the announcement. It is summarized here for completeness and, following the project’s own audit, is explicitly not counted as coverage of the source.

- **Hecke operators** (HeckeOperators.lean, HeckePrime*.lean): the classical coefficient formula $(T_m a)(n) = \sum_{d|\gcd(m,n), (d,N)=1} a\left(\frac{mn}{d^2}\right)$ is defined over an arbitrary additive monoid; linearity, symmetry in m, n , $(T_m a)(1) = a(m)$, multiplicativity on coprime indices, and the prime relations $a(p)^2 = a(p^2) + 1$ (good p) resp. $a(p)^2 = a(p^2)$ ($p \mid N$) are proved.
- **Moonshine** (Moonshine*.lean, MonsterCharacterTable.lean, MonsterIrreps.lean): the 194×194 Monster character table is embedded as data and parsed inside Lean; the McKay–Thompson head-character computation reproduces the first four coefficients 196884, 21493760, 864299970, 20245856256 of $J = j - 744$ from the graded traces (MoonshineBridge.bridge_dim_head) of the first head modules.
- **Compression and codecs** (CharTableCompression.lean, TransitionCompression.lean, Autoencoder*.lean): exact linear-algebra statements about storing the character table—e.g. that conjugation symmetry leaves exactly 172 rows and 172 columns to store, that the coboundary map $c \mapsto (c_j - c_i)$ has kernel exactly the constants (of rank 15) so that a difference

code reconstructs from one origin row, and a rank argument showing that a linear autoencoder reconstructing all 194 irreducible dimension vectors needs hidden width *exactly* 15: width $h \geq 15$ is forced (`MonsterAutoencoder.hidden_layer_lower_bound`) and 15 suffices.

- **Higher-dimensional hyperbolic analogues** (`HigherDimensional.lean`): the radial eigenvalue relation $\lambda = \left(\frac{n-1}{2}\right)^2 + r^2$ on $\mathbb{H}^n$, specializing to $\frac{1}{4} + r^2$ at $n = 2$.
- **Classical background** (`HopfNoether.lean`): the Hopf antipode identity and the ideal form of the Hilbert basis theorem, used as sanity anchors for the umbrella interface `AllTheories.lean`.

9 Faithfulness, scope and limitations

We restate, in this paper’s own voice, the audit carried out in the repository’s `FORMALIZATION_COVERAGE.md`. “Formalized” means a corresponding Lean declaration exists and is proved without `sorry`. It does *not* mean that a numerical assertion copied from the announcement has been independently established.

Target	Faithful coverage
Elementary displayed setup and algebraic identities	$\approx 60\text{--}70\%$
Algorithm descriptions, at the level of abstract/finite cores	$\approx 35\text{--}45\%$
Algorithms as executable certified implementations	$< 10\%$
Announced metadata constants and their arithmetic consequences	$\approx 80\text{--}90\%$ transcribed; 0% independently validated
The 35,416 records, their coefficients, errors and portraits	effectively 0%
Theorems and implementations of the cited bibliography	effectively 0%
Whole announcement, weighted by substantive claims	$\approx 25\text{--}35\%$ as a formal skeleton

Table 2: Coverage estimate. These are audit judgements, not machine-derived metrics; they deliberately separate transcription, proof, implementation and external validation.

Principal gaps.

1. *No analytic Bessel theory.* The distinguished decaying solution K_{ir} is not constructed; its integral representation, decay and truncation estimates are absent. Theorem 3.4 is conditional on being handed a solution of the ODE.
2. *No Selberg trace formula.* There is no equality with a geometric side, no test-function transform pair, no truncation analysis, no class-number machinery. Theorem 6.4 is a valid abstraction of “a localized weighted sum isolates spectral data”, nothing more.
3. *No spectral theorem for the automorphic Laplacian.* The quasimode results (Theorems 6.6–6.7) live in a finite model with an assumed eigenbasis; continuous spectrum, multiplicities, and the boundary estimates of the quasimode literature are absent.
4. *No executable certified Hejhal solver.* The pullback is noncomputable; the Bessel-weighted linear system, the interval matrix solve and all concrete truncation and rounding bounds are absent; contractivity is a hypothesis.
5. *No data.* The announced snapshot, its Arb certificates, its 35,416,000 coefficient intervals and its portraits are not in the repository. Theorems about the metadata are theorems about the copied constants.

6. *No quotient geometry.* A strict geometric fundamental region with measurable structure and explicit boundary identifications, and a quotient-space Laplacian domain, are not built; Theorems 4.1 and 4.2 are the available substitutes.
7. *Trusted base.* (*)-marked results additionally rest on `Lean.ofReduceBool` and `Lean.trustCompiler` (Remark 2.1).

What is genuinely new relative to the announcement. The cusp theory of Section 5 and the combinatorics of Section 7 go beyond what the announcement states: the exact width formula with its full invariance package, the equivalence of the two formulations of cuspidality, the complete squarefree-level classification, and the duality/orbit analysis of the 231 level-tagged classes are proved here in machine-checked form.

10 Reproducibility

The library is a standard Lake project. With Lean 4.28.0 and the pinned Mathlib revision available:

```
lake exe cache get
lake build
```

builds all 82 modules (8108 jobs) with 0 errors and 0 `sorry`; the build emits only style warnings. This paper is built from `paper/paper.tex` with `tectonic paper.tex` (or `latexmk -pdf`); all twelve figures are TikZ/PGFPlots sources inside that file, so the document has no external image dependencies.

To audit a claim: locate the declaration name quoted next to the statement, open the module named in brackets, and check the axiom footprint with `#print axioms` on that declaration.

11 Conclusion

The development described here is best characterized as a *sorry-free formal companion* to a database announcement: it fixes the definitions the announcement uses, proves the elementary analytic identities behind its Fourier expansion, develops in depth the cusp theory that the announcement takes for granted, models the mechanisms of its three certification algorithms in finite or abstract form, and settles exactly the combinatorics of cusps over the announced level set. It is not a reproduction of the announced computation, and Section 9 says precisely why.

The most incremental route to genuine reproduction is, in order: an analytic K_{ir} with decay and truncation bounds; a strict fundamental region with boundary identifications and a quotient Laplacian domain; then a single certification path—most plausibly Hejhal—carried end to end with interval arithmetic, a certified solve, and a concrete stopping theorem; and only then the ingestion of a small, auditable sample of real records with provenance.

A Module index

Module group	Modules
Analytic core	Main, ModifiedBesselODE, FiniteEigenExpansion, InfiniteEigenExpansion, HigherDimensional, Projections

Module group	Modules
Automorphy & domains	AutomorphicSetting, FundamentalDomain, StandardFundamentalDomain, GeneralLevelFundamentalDomain, Gamma0OrbitQuotient, CuspidalityAtCusps
Cusps & widths	CuspWidth, CuspWidthClassification, CuspWidthExamples, CuspScalingConstruction, CuspRepresentativeChange, CuspEquivalence, CuspCuspidalityEquivalence, CuspCuspidalityMinimal, CuspIntegerTranslation, LevelOneCuspidality, CuspClasses, RationalCuspClasses, PrimeLevelCuspWidths, PrimeLevelCuspClasses, SquarefreeLevelCuspWidths, SquarefreeLevelCuspClasses
Certification cores	TraceFormulaIsolation, QuasimodeCertification, QuasimodeConcentration, HejhalHorocycle, HejhalIteration, HejhalCertification, HejhalPullback, IntervalArithmetic, IntervalAggregation, CertifiedComputation, SpectralCertification, FrickeSignCertification, SymmetrySamplingPED
Database arithmetic	DatabaseMetadata, DatabaseRow, DatabaseCuspAccounting, DatabaseCuspDistribution, DatabaseCuspWidthStatistics, DatabaseCuspWidthSupport, DatabaseCuspWidthDuality, DatabaseCuspWidthOrbits, DatabaseCuspWidthOrbitQuotient, DatabaseCuspWidthOrbitStructure, DatabaseCuspWidthOrbitDecomposition
Hecke theory	HeckeOperators, HeckePrimeRelations, HeckePrimePowers
Moonshine & finite groups	Moonshine, MoonshineBridge, MoonshineForms, MoonshineCusps, MoonshineArtifacts, PenumbralMoonshine, RamanujanMockTheta, MonsterCharacterTable, MonsterIrreps, Sporadic, Shapes, HistoricalCityRoutes
Compression & codecs	CharTableCompression, CharTableExplained, CharTableFlatTable, CharTableFullFlatTable, CharTableImaginaryFields, CharTableSerializedCodec, CharTableSymmetries, Transitions, TransitionCompression, TransitionCompression2, Autoencoder, AutoencoderFactored, AutoencoderSymmetry
Umbrella & misc.	AllTheories, HopfNoether

The umbrella module **AllTheories** bundles representative results from every group into a single interface, so that a single import exercises the whole library.

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